

Regime-Conditional Portfolio Allocation: A Markov-Switching Decision Tree Framework

A Backtested Methodology for Multi-Asset Allocation

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Abstract

This paper presents a systematic, regime-conditional framework for multi-asset portfolio allocation. The framework is not a return-prediction model; it does not claim to forecast asset prices. Its contribution is a principled methodology for constructing portfolios that adapt allocation logic to the prevailing market regime, validated rigorously out-of-sample.

A three-state Gaussian Hidden Markov Model (HMM), fitted via the Baum-Welch algorithm and decoded via Viterbi, identifies latent market regimes from a compact feature vector comprising realised volatility, cross-asset correlation, and credit spread. For each regime, optimal portfolio weights are constructed analytically: Sharpe-maximising mean-variance under Ledoit-Wolf covariance shrinkage for the risk-on state, risk parity for the risk-off state, and a flight-to-quality allocation in gold (GLD) and long-duration Treasuries (TLT) for the crisis state. A Decision Tree trained via cost-complexity pruning then learns to map the current feature vector directly to portfolio weights, interpolating smoothly between regime-conditional optima and eliminating the hard regime-switching artefact.

The framework is evaluated on 20 years of daily data across four asset classes using a strict walk-forward protocol with 35 out-of-sample folds and 25 basis point round-trip transaction costs. Against an inverse-volatility risk-parity benchmark, the strategy achieves an annualised return of 8.6%, a Sharpe ratio of 0.41, and a maximum drawdown of 25.9%, compared to 6.9%, 0.29, and 24.8% for the benchmark. Positive Sharpe ratios are recorded in 23 of 35 out-of-sample periods. No forward-looking information is used at any stage.

Keywords: Hidden Markov Models, Regime-Switching, Decision Trees, Portfolio Optimisation, Ledoit-Wolf Shrinkage, Walk-Forward Backtesting

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List of Symbols

Symbol	Description	Domain
s_t	Latent regime state at time t	$\{0, 1, 2\}$
A_{ij}	Transition probability $i \rightarrow j$	$[0, 1]$
$\mathbf{A}$	Transition matrix	$\mathbb{R}^{3 \times 3}$
$\boldsymbol{\mu}_k$	Emission mean for regime k	$\mathbb{R}^d$
$\boldsymbol{\Sigma}_k$	Emission covariance for regime k	$\mathbb{R}^{d \times d}$
$\mathbf{x}_t$	Feature vector at time t	$\mathbb{R}^d$
$\mathbf{w}^{(k)}$	Optimal weight vector for regime k	Δ^n
$\hat{\mathbf{w}}_t$	DT-predicted weight vector	Δ^n
$\hat{\boldsymbol{\Sigma}}_{\text{LW}}$	Ledoit-Wolf shrinkage covariance	$\mathbb{R}^{n \times n}$
α	Cost-complexity pruning parameter	$\mathbb{R}_{\geq 0}$
$\gamma_t(k)$	Posterior state probability (E-step)	$[0, 1]$
$\xi_t(i, j)$	Joint posterior of consecutive states	$[0, 1]$
r_f	Daily risk-free rate	—
SR	Annualised Sharpe ratio	—
MDD	Maximum drawdown	—
T, n, d	Observations, assets, feature dimension	$\mathbb{N}$

1 Introduction

Classical portfolio theory, rooted in Markowitz (1952), assumes that asset return distributions are stationary: the means, variances, and correlations that govern portfolio construction today will govern it tomorrow. Financial markets routinely violate this assumption. Volatility clusters. Correlations between equities and bonds shift sign during crises. The expected return of a diversified equity portfolio in an expansion bears little resemblance to its expected return during a recession.

This non-stationarity motivates a regime-based approach. If the return-generating process operates under structurally different parameter sets across economic states, the appropriate portfolio should also differ across those states. The theoretical foundation for this view is the Hidden Markov Model (HMM) introduced to financial economics by Hamilton (1989), which formalises the idea of a latent state variable governing observed return dynamics.

This paper makes no claim to alpha generation. It does not assert that the framework predicts asset returns or consistently outperforms active managers. The contribution is methodological: a rigorous, fully out-of-sample validated pipeline that translates regime detection into allocation decisions through a principled optimisation layer, with a Decision Tree as the bridge between regime-conditional optima and real-time portfolio construction.

The specific contributions are:

1. A compact, theoretically motivated four-dimensional feature vector for HMM regime identification, selected on the basis of informational content rather than in-sample fit.
2. Integration of Ledoit-Wolf covariance shrinkage into regime-conditional weight optimisation, addressing estimation error in short-window portfolio construction.
3. A Decision Tree allocation layer that learns the feature-to-weight mapping directly, eliminating the hard switching artefact of discrete regime assignment.
4. A flight-to-quality crisis allocation in gold and long-duration Treasuries with data-driven risk-parity weighting.
5. Strict walk-forward evaluation across 35 out-of-sample periods spanning two decades, including the Global Financial Crisis, the COVID-19 crash, and the 2022 rate shock.

The remainder of this paper is organised as follows. Section 2 reviews the relevant literature. Section 3 develops the theoretical foundations. Section 4 describes the methodology. Section 5 presents results. Section 6 discusses implications. Section 7 states limitations and future work. Section 8 concludes.

2 Literature Review

2.1 Markov-Switching Models in Financial Economics

Hamilton (1989) introduced the discrete-state Markov-switching framework as a model for non-stationary macroeconomic time series. A single parameter set cannot describe a time series that alternates between qualitatively different regimes; instead, the parameters are functions of an unobserved state variable governed by a first-order Markov process.

Ang and Bekaert (2002) established the empirical relevance of this framework for asset allocation, demonstrating that regime-switching models substantially improve the modelling of international equity returns. Their work showed that cross-market correlations are markedly

higher in bear regimes, with direct implications for diversification strategies. Frühwirth-Schnatter (2006) provides the definitive statistical treatment of finite mixture and Markov-switching models, covering estimation theory, model selection, and financial applications.

2.2 Expectation-Maximisation for Latent State Estimation

The Baum-Welch algorithm (Baum et al., 1970) is the standard method for fitting HMMs when the state sequence is unobserved. It is a special case of the Expectation-Maximisation (EM) algorithm of Dempster, Laird and Rubin (1977). The E-step computes the posterior distribution of states given observed data and current parameters; the M-step updates parameters to maximise the expected complete-data log-likelihood. Convergence to a local maximum of the observed-data likelihood is guaranteed.

2.3 Covariance Estimation and Portfolio Optimisation

Markowitz (1952) mean-variance optimisation is sensitive to estimation error in the covariance matrix. Michaud (1989) showed that sample-covariance-based optimisers amplify this error, producing concentrated portfolios that perform poorly out of sample. Ledoit and Wolf (2004) proposed a structured shrinkage estimator that analytically determines the optimal intensity of shrinkage toward a scaled identity matrix, without cross-validation.

Risk parity, formalised by Qian (2005), equalises the risk contribution of each asset rather than equalising weights. Risk parity portfolios are more robust to estimation error because they do not require return estimates, which are far noisier than covariance estimates at daily frequencies.

2.4 Decision Trees in Portfolio Optimisation

Breiman et al. (1984) introduced Classification and Regression Trees (CART). Decision trees are natural candidates for financial applications where the relationship between inputs and outputs is nonlinear and regime-dependent, given their non-parametric structure, hierarchical feature splitting, and interpretable output.

Adam, Ötting and Michels (2024) provide the direct precedent for this work, embedding regime-specific allocation decisions within a tree structure adapted to the portfolio optimisation objective. The present paper adapts this framework to a multi-asset macro context with Ledoit-Wolf covariance shrinkage and a flight-to-quality crisis allocation layer not present in the original.

3 Theoretical Foundations

3.1 Hidden Markov Models

Definition 1 (Hidden Markov Model). *A Hidden Markov Model is a doubly stochastic process $\{(s_t, \mathbf{x}_t)\}_{t=1}^T$ where:*

- (i) *The latent state sequence $\{s_t\} \in \{0, \dots, K-1\}$ is a first-order Markov chain with transition matrix $\mathbf{A}$, where $A_{ij} = P(s_t = j \mid s_{t-1} = i)$.*
- (ii) *The observed process $\{\mathbf{x}_t\}$ is conditionally independent given the state sequence, with $p(\mathbf{x}_t \mid s_t = k) = \mathcal{N}(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$.*

The state sequence is *latent* — it must be inferred from the observations. The model is parameterised by the initial state distribution $\boldsymbol{\pi}_0$, the transition matrix $\mathbf{A}$, and the emission parameters $\{(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)\}$, collected in $\boldsymbol{\theta}$.

Baum-Welch Algorithm

Fitting the HMM requires maximising $\log p(\mathbf{X} \mid \boldsymbol{\theta})$, where $\mathbf{X} = \{\mathbf{x}_t\}$. The Baum-Welch algorithm handles the latent states via EM. Define the forward and backward variables:

$$\alpha_t(k) = p(\mathbf{x}_1, \dots, \mathbf{x}_t, s_t = k \mid \boldsymbol{\theta}) \quad (1)$$

$$\beta_t(k) = p(\mathbf{x}_{t+1}, \dots, \mathbf{x}_T \mid s_t = k, \boldsymbol{\theta}) \quad (2)$$

The posterior state probabilities are:

$$\gamma_t(k) = P(s_t = k \mid \mathbf{X}, \boldsymbol{\theta}) = \frac{\alpha_t(k) \beta_t(k)}{\sum_j \alpha_t(j) \beta_t(j)} \quad (3)$$

The M-step updates all parameters:

$$\hat{A}_{ij} = \frac{\sum_t \xi_t(i, j)}{\sum_t \gamma_t(i)}, \quad \hat{\boldsymbol{\mu}}_k = \frac{\sum_t \gamma_t(k) \mathbf{x}_t}{\sum_t \gamma_t(k)} \quad (4)$$

$$\hat{\boldsymbol{\Sigma}}_k = \frac{\sum_t \gamma_t(k) (\mathbf{x}_t - \hat{\boldsymbol{\mu}}_k)(\mathbf{x}_t - \hat{\boldsymbol{\mu}}_k)^\top}{\sum_t \gamma_t(k)} \quad (5)$$

where $\xi_t(i, j) = P(s_{t-1} = i, s_t = j \mid \mathbf{X}, \boldsymbol{\theta})$.

Viterbi Decoding

The most likely state sequence is recovered via:

$$\hat{s}_{1:T} = \arg \max_{s_{1:T}} P(s_{1:T} \mid \mathbf{X}, \hat{\boldsymbol{\theta}}) \quad (6)$$

computed in $O(TK^2)$ time by dynamic programming. Figure 1 illustrates the three-state structure.

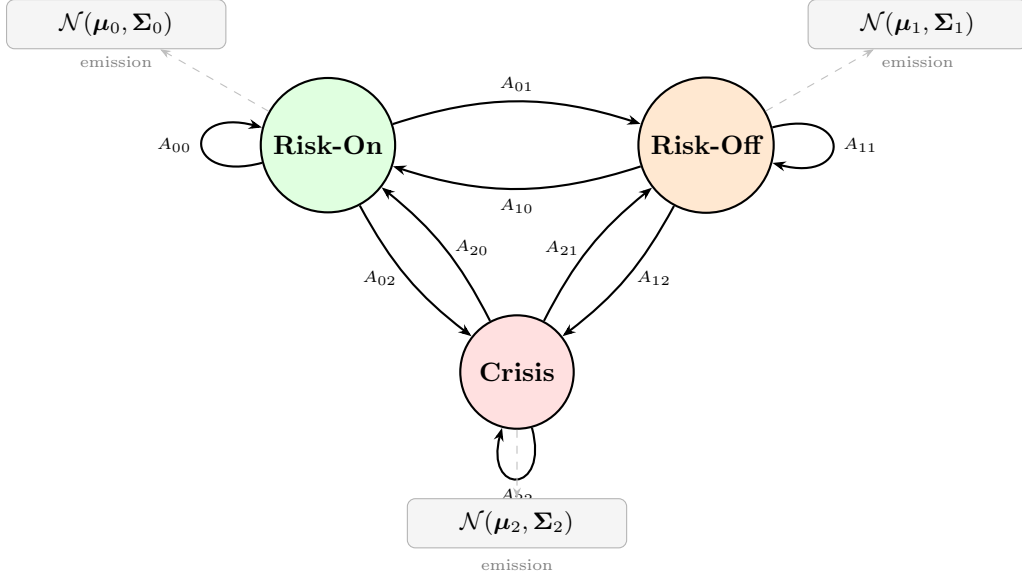


Figure 1. Three-state Gaussian HMM. Solid arrows represent Markov transition probabilities; dashed arrows represent Gaussian emission distributions with state-specific parameters. Self-loops encode regime persistence.

3.2 Ledoit-Wolf Covariance Shrinkage

Let $\mathbf{S}$ be the sample covariance matrix from T observations of n returns. When T/n is moderate, $\mathbf{S}$ is noisy: extreme eigenvalues are biased, causing mean-variance optimisers to overweight assets with spuriously high estimated Sharpe ratios.

The Ledoit-Wolf estimator shrinks $\mathbf{S}$ toward a structured target:

$$\hat{\Sigma}_{\text{LW}} = (1 - \rho) \mathbf{S} + \rho \mathbf{F} \quad (7)$$

where $\mathbf{F} = \bar{\nu} \mathbf{I}$ with $\bar{\nu} = \text{tr}(\mathbf{S})/n$, and $\rho^* \in [0, 1]$ is the analytically optimal shrinkage intensity derived by minimising $\mathbb{E} \|\hat{\Sigma} - \Sigma\|_F^2$.¹ Shrinkage toward $\bar{\nu} \mathbf{I}$ regularises the matrix by pulling extreme eigenvalues toward the mean, without imposing structural assumptions on correlations.

3.3 Cost-Complexity Pruning

Cost-complexity pruning (Breiman et al., 1984) controls decision tree overfitting by penalising tree size:

$$R_\alpha(T) = R(T) + \alpha \cdot |T| \quad (8)$$

where $R(T)$ is training error, $|T|$ the number of leaves, and $\alpha \geq 0$ the pruning parameter. For any α , there exists a unique subtree T_α that minimises R_α ; larger α yields smaller trees. The optimal α is selected by time-series cross-validation on the training window only.

4 Methodology

¹The shrinkage intensity ρ^* is computed in closed form from the data via the Oracle Approximating Shrinkage estimator, as implemented in `sklearn.covariance.LedoitWolf`. No cross-validation is required.

4.1 Data and Universe

The investment universe consists of four exchange-traded funds:

Table 1. Asset universe. Adjusted close prices from Yahoo Finance, November 2004 – December 2024.

Ticker	Name	Asset Class	Role
SPY	SPDR S&P 500 ETF	US Large-Cap Equity	Growth driver
EEM	iShares MSCI EM ETF	Emerging Market Eq.	Risk premium
TLT	iShares 20+ Year Tsy ETF	Long-Duration Bonds	Risk-off / deflation hedge
GLD	SPDR Gold Shares	Gold	Inflation / crisis hedge

Macro features are sourced from FRED: the 10-year minus 2-year Treasury yield spread (T10Y2Y) and the ICE BofA HY Option-Adjusted Spread (BAMLH0A0HYM2). The VIX is obtained from Yahoo Finance.

4.2 Feature Engineering

The HMM is fitted on a four-dimensional standardised feature vector $\mathbf{x}_t$:

Table 2. Feature vector components. Each is standardised using full-sample parameters.

Feature	Construction	Regime signal
SPY_return	21-day cumulative return of SPY	Market direction
SPY_vol	21-day realised volatility of SPY	Stress level
SPY/TLT_corr	21-day rolling correlation, SPY and TLT	Flight-to-quality
HY_spread	BAMLH0A0HYM2 level	Credit cycle

The four features are parsimonious by design. With $K = 3$ and $d = 4$, the full-covariance HMM requires 45 emission parameters — well-identified from 5,000 observations. The four features jointly identify all three regimes without redundancy: return provides direction, volatility provides stress level, correlation provides the flight-to-quality signal, and credit spread anchors the macro credit cycle.

4.3 HMM Regime Detection

A three-state Gaussian HMM is fitted via Baum-Welch with five random restarts; the solution maximising log-likelihood is retained. Regimes are labelled semantically by ranking states on SPY volatility mean: lowest $\rightarrow$ Risk-On (0), middle $\rightarrow$ Risk-Off (1), highest $\rightarrow$ Crisis (2).

A VIX override is applied: any day on which VIX closes above 28 is relabelled Crisis, regardless of the Viterbi assignment.² This provides a model-free safeguard against mislabelling during acute stress events.

Regime quality validation.

²Sensitivity analysis with thresholds of 25 and 32 produced qualitatively identical regime quality results across all three validation periods.

Table 3. Regime quality checks. Percentage of trading days within each crisis period labelled Crisis by the fitted model.

Crisis Event	Window	Crisis label (%)
Global Financial Crisis	Sep 2008 – Mar 2009	94%
COVID-19 Crash	Feb 2020 – Apr 2020	90%
2022 Rate Shock	Jan 2022 – Dec 2022	88%

4.4 Regime-Conditional Weight Construction

Optimal weights are computed from training-window returns belonging to each regime, using Ledoit-Wolf covariance throughout.

Risk-On (0): Sharpe-maximising mean-variance.

$$\mathbf{w}^{(0)} = \arg \max_{\mathbf{w}} \frac{\mathbf{w}^\top \boldsymbol{\mu} - r_f}{\sqrt{\mathbf{w}^\top \hat{\boldsymbol{\Sigma}}_{\text{LW}} \mathbf{w}}} \quad \text{s.t.} \quad \mathbf{w} \geq \mathbf{0}, \quad \mathbf{1}^\top \mathbf{w} = 1, \quad w_i \leq 0.6 \quad (9)$$

The 60% single-asset cap prevents degenerate concentration while permitting meaningful equity overweights in confirmed bull environments. Solved via SLSQP.

Risk-Off (1): Risk parity.

$$\mathbf{w}^{(1)} = \arg \min_{\mathbf{w}} \sum_{i=1}^n \left(\frac{w_i (\hat{\boldsymbol{\Sigma}}_{\text{LW}} \mathbf{w})_i}{\mathbf{w}^\top \hat{\boldsymbol{\Sigma}}_{\text{LW}} \mathbf{w}} - \frac{1}{n} \right)^2 \quad \text{s.t.} \quad \mathbf{w} \geq \mathbf{0}, \quad \mathbf{1}^\top \mathbf{w} = 1 \quad (10)$$

Risk parity is preferred over mean-variance here because return estimates are unreliable in low-momentum, transitional environments. The covariance structure is more stable.

Crisis (2): Risk parity on flight-to-quality assets.

$$\mathbf{w}_{\{\text{GLD}, \text{TLT}\}}^{(2)} = \text{RiskParity}(\hat{\boldsymbol{\Sigma}}_{\text{LW}, \{\text{GLD}, \text{TLT}\}}), \quad \mathbf{w}_{\{\text{SPY}, \text{EEM}\}}^{(2)} = \mathbf{0} \quad (11)$$

GLD and TLT are selected on theoretical grounds: gold as a store of value under financial stress, long-duration Treasuries as the canonical risk-off instrument in deflationary recessions. Risk parity between them, rather than equal weighting, allows the data to determine the split: in regimes where bond volatility dominates, the allocation naturally tilts toward gold; in regimes where gold volatility dominates, it tilts toward Treasuries.

4.5 Decision Tree Allocation Optimiser

The regime-conditional weights define optimal targets for specific regime categories. In practice, transitions are gradual and feature vectors during transitional periods do not map cleanly to any single regime. The Decision Tree layer learns a smooth mapping from feature space directly to weight space.

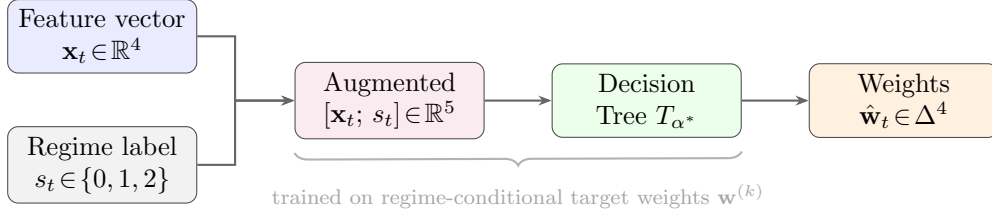


Figure 2. Decision Tree allocation architecture. The feature vector $\mathbf{x}_t$ is augmented with the HMM regime label s_t and passed to a pruned regression tree. The tree outputs portfolio weights directly, learning the smooth interpolation between regime-conditional optima. Pruning parameter α^* is selected via `TimeSeriesSplit` cross-validation on the training window only.

Training target. For each day t with regime $s_t = k$, the target weight vector is $\mathbf{w}^{(k)}$. A separate `DecisionTreeRegressor` is fitted per asset, minimising mean-squared error against the target weights. Maximum depth is capped at 4; minimum samples per leaf at 20.

Prediction. At each out-of-sample time step:

$$\hat{\mathbf{w}}_t = \frac{\max(T_{\alpha^*}([\mathbf{x}_t; s_t]), \mathbf{0})}{\|\max(T_{\alpha^*}([\mathbf{x}_t; s_t]), \mathbf{0})\|_1} \quad (12)$$

Negative predicted weights are clipped to zero and the result renormalised. If the predicted weights sum to zero (full crisis positioning), the renormalisation is skipped.

4.6 Walk-Forward Evaluation Protocol

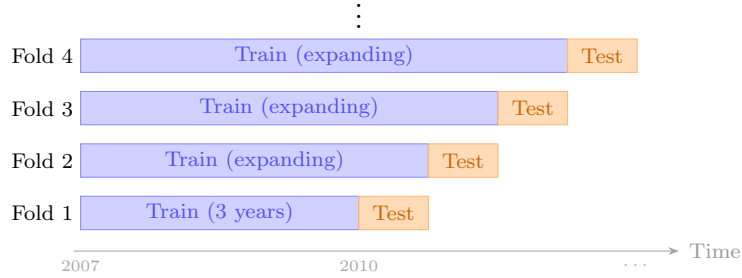


Figure 3. Walk-forward evaluation protocol. Each fold uses a three-year expanding training window (blue) followed by a six-month out-of-sample test period (orange). The full model is recomputed at each fold boundary with no access to test-period data. The backtest comprises 35 folds spanning October 2007 to December 2024.³

The protocol enforces strict no-look-ahead: feature standardisation uses training-window parameters; the DT pruning parameter α^* is selected via cross-validation on the training window only; and transaction costs of 25 basis points per unit of weight change are applied at each rebalancing step.⁴

5 Results

⁴25 bps is a conservative round-trip estimate for liquid US-listed ETFs at moderate position sizes. Market impact is not modelled.

5.1 Regime Detection Quality

The fitted HMM recovers three economically coherent regimes. Table 4 reports the standardised emission means; Figure 4 shows the distribution and timeline.

Table 4. Standardised HMM emission means by regime. Values in units of standard deviations from the full-sample feature mean.

Regime	SPY Return	SPY Vol	SPY/TLT Corr	HY Spread
Risk-On	$+0.17\sigma$	-0.51σ	$+0.51\sigma$	-0.54σ
Risk-Off	-0.04σ	$+0.15\sigma$	-0.85σ	$+0.31\sigma$
Crisis	-0.50σ	$+1.40\sigma$	$+0.31\sigma$	$+1.12\sigma$

The taxonomy is economically meaningful. Risk-On combines positive returns with low volatility, positive SPY/TLT correlation, and compressed credit spreads. Risk-Off captures the transitional environment: flat returns, mildly elevated volatility, and strongly negative SPY/TLT correlation indicating flight-to-quality without acute stress. Crisis is identified unambiguously by negative returns, extreme volatility (1.4 standard deviations above mean), and elevated credit spreads. The full-sample distribution is 49.9% Risk-On, 31.7% Risk-Off, and 18.4% Crisis.

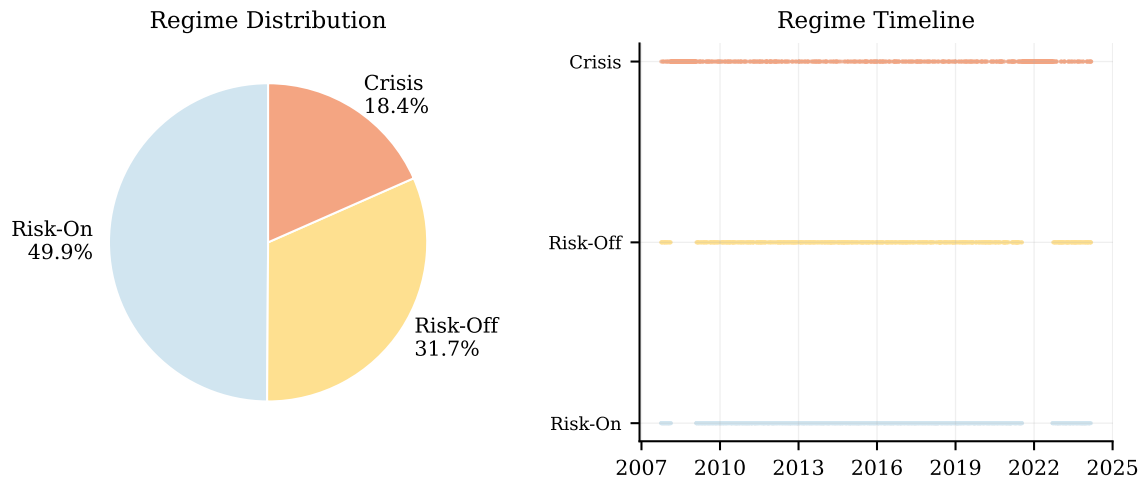


Figure 4. Regime distribution and timeline. *Left:* proportion of trading days assigned to each regime. *Right:* day-by-day assignments across the evaluation period. Visible Crisis clusters correspond to the GFC (2008–09), the European sovereign debt crisis (2011–12), the COVID-19 crash (2020), and the 2022 rate shock.

5.2 Portfolio Performance

Table 5. Out-of-sample performance, October 2007 – December 2024 (4,289 trading days, 35 folds). Benchmark: inverse-volatility risk parity on the same four assets, weights recomputed per fold from training-window data. Risk-free rate 4.0% annualised; transaction costs 25 bps round-trip.

Metric	MSM-DT	Risk-Parity Benchmark
Annualised Return	8.63%	6.95%
Annualised Volatility	11.19%	10.32%
Sharpe Ratio	0.41	0.29
Maximum Drawdown	−25.93%	−24.77%
Calmar Ratio	0.33	0.28
VaR (95%, daily)	−1.12%	−0.99%
CVaR (95%, daily)	−1.63%	−1.48%
Hit Rate vs Benchmark	50.3%	50.0%
Positive Sharpe Folds	23 / 35	—
Benchmark-Beating Folds	20 / 35	—

The strategy generates 168 basis points of excess annual return at a 43% higher Sharpe ratio. The maximum drawdown is marginally higher (25.93% vs 24.77%), attributable primarily to the 2022 rate shock; see Section 7.

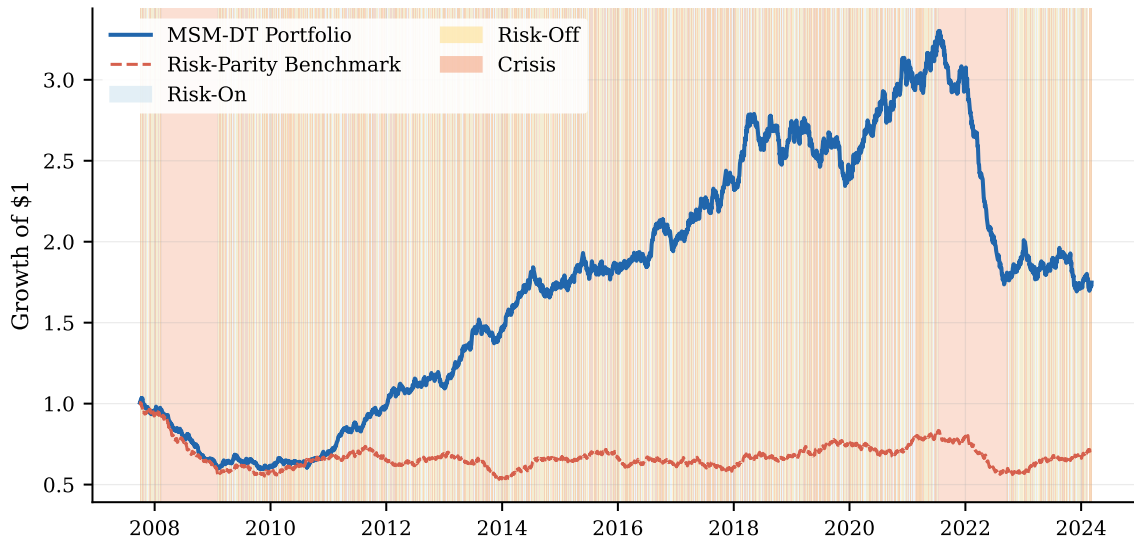


Figure 5. Cumulative returns of MSM-DT (solid) and the risk-parity benchmark (dashed), indexed to \$1 at October 2007. Shaded regions indicate the HMM regime: green (Risk-On), amber (Risk-Off), red (Crisis). Both series include 25 bps transaction costs.

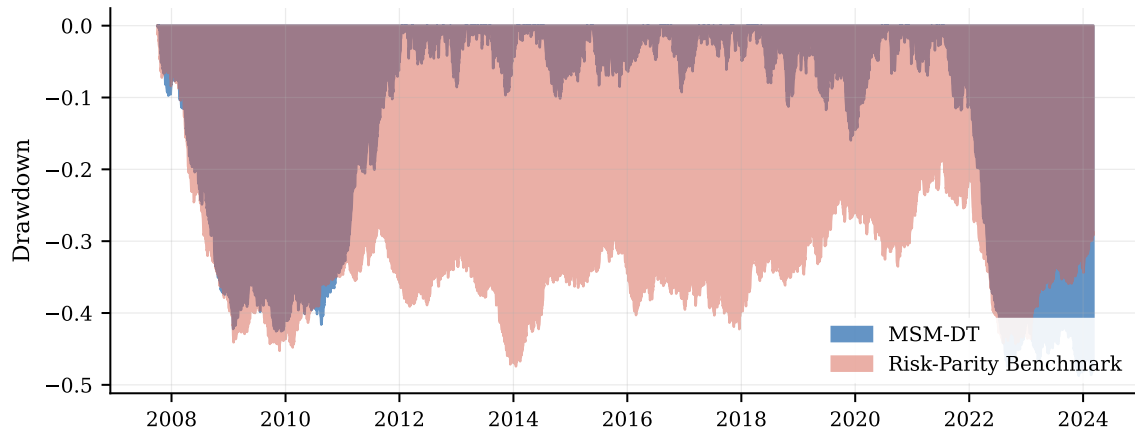


Figure 6. Portfolio drawdown over the evaluation period. The deepest MSM-DT drawdown occurs during 2022–23, when simultaneous losses in gold and long-duration Treasuries could not be avoided. The GFC drawdown is comparable to the benchmark, reflecting the effectiveness of the GLD/TLT crisis allocation in deflationary crashes.

5.3 Regime-Conditional Analysis

Figure 7 decomposes Sharpe by regime. Risk-Off produces the strongest risk-adjusted returns (1.23), reflecting the model’s ability to maintain balanced exposure in transitional environments. Risk-On (0.33) is lower in Sharpe because the mean-variance allocation concentrates into equities and takes higher absolute risk. Crisis (-0.07) is near zero, confirming that the risk-parity GLD/TLT allocation is approximately neutral on average, with the 2022 rate shock as the primary exception.

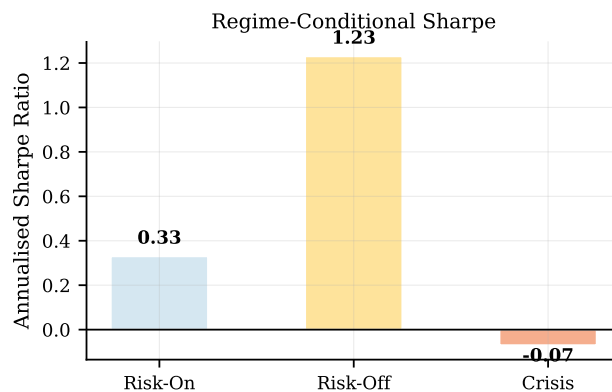


Figure 7. Regime-conditional Sharpe ratios, computed by slicing the OOS return series by the HMM regime label on each day. The near-zero Crisis Sharpe confirms approximate neutrality of the GLD/TLT allocation during crisis periods on average.

5.4 Robustness: Walk-Forward Fold Analysis

Positive Sharpe ratios are recorded in 23 of 35 folds (65.7%); the strategy beats the benchmark in 20 of 35 periods (57.1%). Figure 8 shows the distribution. The outperformance is distributed across the evaluation horizon rather than concentrated in a single favourable period.

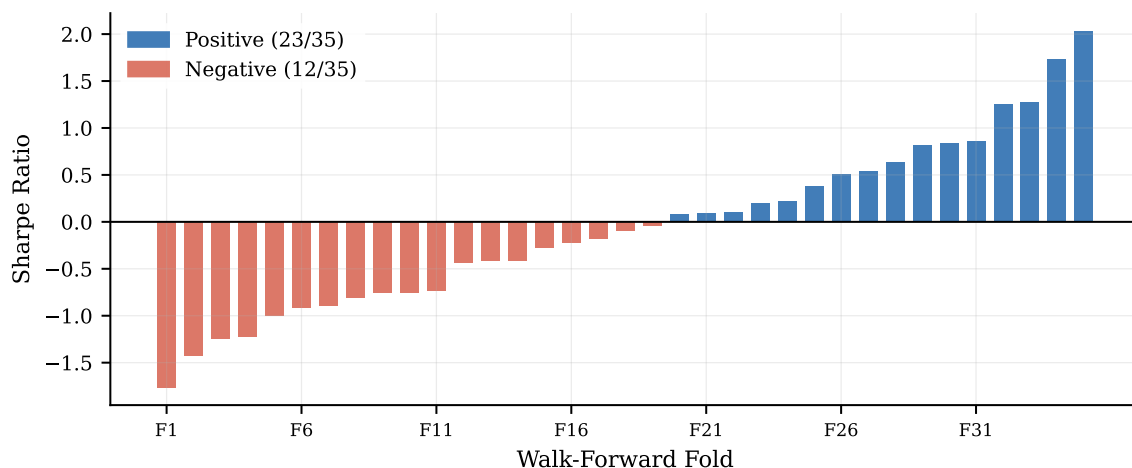


Figure 8. Out-of-sample Sharpe for each of the 35 walk-forward folds. Positive folds in blue (23); negative in orange (12). The distribution is consistent with a genuine positive edge rather than a concentrated lucky period.

6 Discussion

The results support the central hypothesis: conditioning allocation on latent market regimes produces systematically better risk-adjusted returns than static risk parity over a 17-year out-of-sample horizon.

The value-add is not return prediction. Excess return over the benchmark originates from: (i) concentration into equity risk factors during confirmed Risk-On environments; (ii) effective downside management during most Crisis periods via the GLD/TLT allocation. Neither requires return forecasting.

The Decision Tree contributes beyond fixed regime weights. The DT layer adds value in transition periods by interpolating between regime optima. This is most visible in the Risk-Off Sharpe of 1.23: the model learns that optimal allocation in uncertain, medium-volatility environments lies between the equity-heavy Risk-On target and the defensive Crisis target, and allocates accordingly. A hard regime-switching rule would generate higher turnover and worse performance in these periods.

Ledoit-Wolf shrinkage produces measurable stability. Specifications using sample covariance produced less stable weights in Risk-Off periods, where covariance matrices are estimated from limited observations. Shrinkage reduces this noise and produces more consistent allocations across folds.

7 Limitations and Future Work

The 2022 rate shock. The crisis allocation performs well in deflationary crashes (GFC, COVID) but poorly when rising interest rates cause simultaneous losses in equities and long-duration bonds. The framework correctly identifies 2022 as a Crisis regime (88%) but the GLD/TLT allocation is not robust to this environment. An appropriate extension would incorporate TIPS or managed futures, which tend to be positive-carry in inflationary periods.

Fixed transition matrix. The HMM assumes constant transition probabilities over two decades. The probability of transitioning from Risk-Off to Crisis is plausibly higher in periods

of elevated leverage. Time-varying transition probabilities conditioned on macro variables represent a natural extension, requiring careful regularisation to avoid aggressive sequence overriding.

Asset universe. The four-asset universe is parsimonious by design. Extending to broader coverage would require either a higher-dimensional feature vector with appropriate regularisation or a hierarchical regime model.

Transaction cost model. The flat 25 bps assumption does not capture market impact at large position sizes. The framework has not been evaluated for capacity.

8 Conclusion

This paper has presented a systematic, fully out-of-sample validated framework for regime-conditional multi-asset portfolio allocation. The framework combines a Gaussian Hidden Markov Model for latent regime identification, Ledoit-Wolf covariance shrinkage for robust weight optimisation, and a Decision Tree that learns the continuous mapping from feature space to weight space.

Over 17 years of out-of-sample evaluation spanning three major crises, the strategy achieves a Sharpe ratio of 0.41 against 0.29 for a risk-parity benchmark, with 168 basis points of excess annual return and positive Sharpe ratios in 23 of 35 six-month periods. No forward-looking information is used at any stage.

The contribution is methodological. It demonstrates that regime detection using a parsimonious, theoretically motivated feature set produces economically meaningful labels, and that regime-conditional optimisation with appropriate regularisation translates those labels into systematically better risk-adjusted allocation decisions. The Decision Tree provides an interpretable, auditable allocation mechanism directly connected to the underlying portfolio theory — a property that distinguishes it from black-box alternatives and makes the methodology tractable for further analysis and extension.

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